Agenda-Setting Beyond Votes: Sectoral Campaign Contributions and Legislative Speech in the U.S. Senate

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Introduction

The role of money in politics has long been a central concern in political science and political economy and, in fact, a large literature examines whether campaign contributions influence legislators' behavior. While early work framed this debate in terms of whether "money buys votes," more recent research emphasizes alternative mechanisms of influence, such as access, agenda-setting, and the allocation of legislative effort. Despite this shift, we still know relatively little about how campaign contributions shape legislators' public communication.

Legislative speech is a central window into political behavior. Unlike roll-call votes, speeches reveal how legislators allocate attention across issues, signal their priorities and interests, and engage in agenda-setting. Advances in text-as-data methods allow us to analyze large corpora of political speech systematically. Yet most studies of congressional speech abstract from the sectoral origins of political influence, while the campaign finance literature rarely treats speech as an outcome of interest.

This paper bridges these two strands by examining whether sector-specific campaign contributions are associated with systematic differences in the thematic composition of U.S. Senate floor speeches. I combine text-as-data methods applied to Senate speeches from 2011 to 2024 with detailed data on campaign contributions disaggregated by economic sector. The unit of analysis is the senator–electoral cycle, which aligns speech data with campaign finance information and allows the analysis to focus on persistent patterns of agenda emphasis rather than short-term rhetorical fluctuations.

The analysis tests three hypotheses. First, campaign contributions from a given economic sector are associated with increased attention to the corresponding topic in legislative speech (H1). Second, legislators' ideological positions structure baseline agenda priorities independently of campaign finance (H2). Third, the relationship between donations and speech is topic-specific rather than diffuse across policy domains (H3).

The results point to three main findings. First, there is partial but meaningful evidence of sector–topic congruence: for several policy domains, higher lagged contributions are associated with greater attention to the corresponding topic in floor speeches. Second, ideological positioning plays an important role in shaping agenda emphasis, with liberal and conservative senators prioritizing different sets of issues even after accounting for campaign finance and institutional factors. Third, the effect of campaign contributions on speech is mostly concentrated along matching sector–topic pairs.

The findings suggest that campaign contributions may influence legislative speech through targeted agenda-setting dynamics. This paper contributes to a more nuanced understanding of how money, ideology, and institutional context jointly shape political behavior in the U.S. Senate.

Literature Review

The increasing availability of large-scale textual data has fostered a rapidly expanding literature using text-as-data methods to study political behavior and representation. Legislative speech has emerged as a particularly valuable source in this context, as it provides rich information on politicians' priorities, ideological positioning, and communication strategies (Grimmer and Stewart 2013; Gentzkow and Shapiro 2010). Unlike roll-call votes, which capture discrete policy choices at specific points in time, speeches allow researchers to observe how legislators allocate attention across issues and publicly frame political debates (Quinn et al. 2010).

Methodological contributions have played a central role in legitimizing the use of political texts as quantitative data. Foundational work emphasizes both the promise and the pitfalls of automated content analysis and highlights the importance of aligning measurement strategies with substantive research questions (Grimmer and Stewart 2013). Within this framework, congressional speeches have been used to extract information about political positions, polarization, and agenda-setting (Gentzkow et al. 2019; Lauderdale and Herzog 2016).

A substantial strand of the literature focuses on recovering ideological positions from legislative speech. These studies typically rely on scaling methods or high-dimensional classification techniques to place legislators along a left–right ideological dimension, often validating text-based measures against roll-call–based ideal points (Lauderdale and Herzog 2016). Prominent contributions show that speech content captures partisan and ideological differences and that polarization in rhetoric has increased over time in ways that closely mirror patterns observed in voting behavior (Gentzkow et al. 2019; Taddy 2013).

Beyond ideological positioning, scholars have used legislative speech to study political attention and agenda-setting. Rather than asking where legislators are located ideologically, this literature examines which issues they choose to emphasize and how attention is distributed across policy domains. A key insight is that the allocation of attention reflects strategic considerations, institutional constraints, and responsiveness to external actors (Quinn et al. 2010; Jones and Baumgartner 2005). Recent work documents long-run changes in rhetorical styles, shifts in the use of evidence versus intuition, and the evolution of partisan language in the U.S. Congress (Tucker 2020; Aroyehun et al. 2025), as well as variation in the emotional intensity of legislative speech (Dietrich et al. 2019). Taken together, this literature establishes legislative speech as a central object of analysis for understanding political behavior and agenda-setting.

A central insight of the literature on legislative behavior is that ideology structures also patterns of political communication. Legislators' ideological positions shape how policy problems are interpreted, which issues are perceived as salient, and how attention is allocated across competing topics (Poole and Rosenthal 1997; Gentzkow et al. 2019). As a result, ideological differences are reflected not only in final policy choices but also in the thematic composition of legislative speech.

A large body of research demonstrates that ideological positioning can be recovered from political texts and that speech content tracks partisan and ideological divides. Studies comparing text-based measures with voting-based ideal points find a strong correspondence between the two, indicating that ideology is embedded in legislative rhetoric (Lauderdale and Herzog 2016). Beyond the recovery of ideological positions, several contributions emphasize that ideology affects which issues legislators choose to emphasize, independently of short-term incentives. Research on agenda-setting shows that political orientations are associated with systematic differences in issue attention, with conservative legislators emphasizing defense, security, and energy, and liberal legislators focusing more heavily on social policy, labor, and healthcare (Jones and Baumgartner 2005). Long-run analyses of congressional rhetoric support this view, showing that changes in issue emphasis align closely with broader trends in ideological polarization rather than with short-term electoral or institutional shocks (Aroyehun et al. 2025; Tucker 2020).

In empirical work on the U.S. Congress, ideological positions are commonly measured using DW-NOMINATE scores, which estimate legislators' ideal points based on roll-call votes (Poole and Rosenthal 1997). DW-NOMINATE has become a standard measure in studies of congressional behavior, capturing differences in policy preferences that extend beyond party affiliation alone (Bateman 2016). Controlling for ideological positioning therefore plays a crucial role in distinguishing baseline agenda priorities from variation potentially associated with other factors, such as campaign finance.

A vast literature in political science examines the role of campaign contributions and lobbying in shaping political outcomes. Early empirical work focused primarily on whether financial support affects legislators' voting behavior, often framing the question in terms of whether "money buys votes." The evidence on this point is mixed, and many studies find limited or unstable effects of campaign contributions on roll-call voting once ideology and party affiliation are taken into account (Ansolabehere et al. 2003). These findings have motivated a reconsideration of the mechanisms through which money influences politics.

Subsequent research has increasingly emphasized alternative channels of influence, shifting attention away from final votes toward access, agenda-setting, and the allocation of legislative effort. From this perspective, campaign contributions are understood less as instruments to directly alter policy choices and more as tools that facilitate access to policymakers and shape which issues receive attention. Classic work shows that organized interests primarily influence politics by mobilizing bias in agenda-setting and committee activity rather than by changing legislators' voting decisions (Hall and Wayman 1990). This perspective has been further developed in the concept of lobbying as legislative subsidy, which emphasizes how organized interests provide informational and organizational resources that reallocate legislators' attention and effort (Hall and Deardorff 2006). Formal models reinforce this interpretation by conceptualizing contributions as a means to secure access and information advantages rather than direct policy concessions (Austen-Smith 1995; Grossman and Helpman 1994).

A growing empirical literature supports the view that access constitutes a central channel of influence. Using a randomized field experiment, Kalla and Broockman show that campaign contributions significantly increase the likelihood that interest groups obtain meetings with congressional offices, even when policy positions remain unchanged (Kalla and Broockman 2016). Related evidence demonstrates that political connections and lobbying activities affect access to policymakers and the allocation of political attention, reinforcing the idea that influence operates upstream of final policy outcomes (Blanes i Vidal et al. 2012; Bertrand et al. 2014).

While the access-based interpretation of campaign finance has gained wide acceptance, relatively little work has examined how this influence manifests itself in legislators' public communication. Legislative speech represents a natural and observable form of political effort, reflecting the issues legislators choose to emphasize publicly and the signals they send to organized interests. From an agenda-setting perspective, speech can therefore be understood as an integral component of political influence rather than as a purely symbolic activity (Jones and Baumgartner 2005; Quinn et al. 2010). Existing studies that consider communication outcomes often focus on narrow policy domains or rely on case-specific designs, limiting their generalizability. Moreover, much of the literature lacks a systematic mapping between the sectoral origins of campaign contributions and the thematic structure of legislative speech. As a result, it remains unclear whether financial support from specific economic sectors is associated with targeted increases in attention to corresponding policy domains, or whether campaign contributions are instead linked to more diffuse changes in rhetorical behavior (Baumgartner et al. 2009; Drutman 2015; Goel et al. 2023).

A small but growing body of work suggests that campaign contributions may affect various forms of legislative effort beyond voting, including bill sponsorship, committee activity, and speechmaking. Sector-specific interests have been shown to shape rhetorical focus in energy-related legislative activity (Iliev and Brandt 2021), and empirical studies show that financial support can influence how legislators allocate time and effort across issues even when final policy outcomes remain unchanged (Bombardini and Trebbi 2011; Grimmer 2013, 2015). More recent work explicitly links campaign finance to patterns of legislative communication, indicating that speechmaking constitutes a meaningful outcome through which financial influence may operate (Parmigiani 2025). Recent causal evidence further shows that firm donations can affect political rhetoric and issue emphasis (Cagé et al. 2024).

Taken together, this literature points to two conclusions. First, legislative speech constitutes a meaningful dimension of political behavior that captures agenda-setting and issue attention (Grimmer and Stewart 2013; Quinn et al. 2010). Second, while the study of campaign finance has increasingly moved away from the question of whether money directly buys votes, there is growing consensus that financial resources shape political outcomes primarily through access, agenda-setting, and the allocation of legislative effort (Hall and Wayman 1990; Kalla and Broockman 2016).

Despite these advances, an important gap remains. Studies of legislative speech tend to focus on ideology, polarization, or broad measures of political attention, often abstracting from the sectoral origins of political influence. Conversely, the literature on contributions and lobbying rarely examines how financial support from specific sectors translates into systematic changes in legislators' public communication. As a result, we still lack a comprehensive understanding of whether and how sector-specific campaign contributions are associated with targeted changes in agenda emphasis across policy domains. This paper addresses this gap by linking sectoral campaign contributions to the thematic structure of legislative speech in the U.S. Senate and by providing systematic evidence on the sectoral specificity of financial influence in agenda-setting.

Based on this framework, the analysis tests three hypotheses.

H1 - Senators who receive higher campaign contributions from a given economic sector devote greater attention to the corresponding policy topic in their speech. This hypothesis follows directly from access- and agenda-based theories of political influence, which predict that organized interests seek to increase attention to their policy domains rather than to directly alter votes (Hall and Wayman 1990; Austen-Smith 1995).

H2 - Legislators' ideological positions systematically structure agenda emphasis in legislative speech. This hypothesis reflects extensive evidence that ideology shapes baseline policy priorities and patterns of political communication (Poole and Rosenthal 1997; Lauderdale and Herzog 2016).

H3 - The association between campaign contributions and legislative speech is topic-specific rather than diffuse, with stronger effects along matching sector–topic pairs than across unrelated domains. This hypothesis speaks to whether campaign finance is linked to targeted agenda influence or to broader, non-specific changes in rhetorical behavior (Baumgartner et al. 2009; Drutman 2015).

Together, these hypotheses allow the paper to clarify how money in politics relates to agenda-setting and political communication, and to provide systematic evidence on the sectoral specificity of financial influence in legislative speech.

Data, Dictionary and Empirical Strategy

Congressional Speech Data

This paper is based on a corpus of floor speeches delivered in the United States Senate between 2011 and 2024. Floor speeches constitute the main source for studying agenda-setting and political communication within the U.S. Congress, because they represent the means through which senators publicly articulate policy preferences and show their positions to voters and organized interests on almost every subject. The corpus includes all substantive speeches delivered during legislative sessions. Procedural remarks and purely technical interventions are excluded to focus on content that reflects senators' policy concerns rather than parliamentary formalities.

The speech dataset is deliberately restricted to the post-2010 period to ensure greater comparability in the campaign finance environment faced by senators. The Supreme Court's decision in Citizens United v. Federal Election Commission substantially changed the regulation of independent political spending and reshaped the broader context in which campaign finance operates. By focusing on speeches from 2011 onward, the analysis avoids combining observations from substantially different regulatory environments and measures the relationship between contributions and speech within a more consistent institutional period.

The resulting dataset covers the universe of senators serving during the period of analysis and spans seven electoral cycles. The unit of analysis is the senator–electoral cycle. All speeches delivered within a given electoral cycle are aggregated into a single document. This aggregation strategy serves two purposes. First, it aligns the temporal structure of the textual data with campaign finance information, which is also organized at the electoral-cycle level. Second, it reduces noise arising from short-term rhetorical fluctuations, media-driven reactions, and idiosyncratic speeches, allowing the analysis to focus on persistent patterns in agenda emphasis rather than transitory responses to specific events.

After aggregation, each senator–cycle observation is associated with a vector of topic salience measures capturing the relative emphasis devoted to different policy domains within the senator's overall speech production during the cycle. These measures constitute the core dependent variables in the empirical analysis and provide a systematic representation of how legislators allocate attention across issues under a common post–Citizens United campaign finance regime.

Campaign Finance Data

Data on campaign contributions are drawn from OpenSecrets, which provides detailed information on political donations to federal candidates disaggregated by economic sector. OpenSecrets aggregates individual and organizational contributions and assigns them to donors' primary economic interests, producing a consistent sectoral classification widely used in the literature on campaign finance and lobbying. Contributions are classified into thirteen broad sectors, including agribusiness, energy, defense, finance, health, labor, transportation, and others. This classification is particularly well suited to the present analysis, as it allows for a direct and substantively meaningful correspondence between sources of financial support and policy-relevant domains discussed in legislative speech.

For each senator and electoral cycle, sector-specific contributions are aggregated at the candidate level and lagged by one cycle. Lagging donations reflects the assumption that campaign contributions precede and potentially shape subsequent communication strategies, rather than responding contemporaneously to speech content. From an empirical perspective, this choice mitigates concerns about simultaneity and reverse causality between donations and legislative speech, while aligning the temporal structure of the campaign finance data with the senator–cycle unit of analysis adopted throughout the paper.

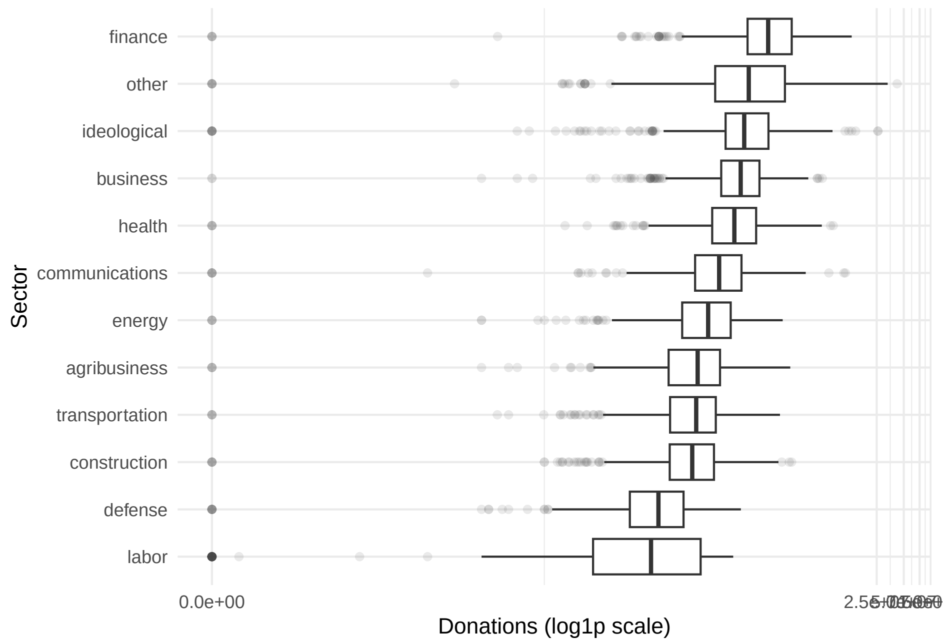

Figure 1: Donations by sector (log scale)

Figure 1 reports the distribution of campaign contributions by sector on a logarithmic scale. The figure highlights substantial heterogeneity in both the magnitude and dispersion of donations across sectors. Contributions from finance, business-related sectors, and ideological organizations tend to be larger on average and exhibit greater dispersion, indicating the presence of both very high and more modest donors. By contrast, sectors such as labor and defense display lower median contribution levels and more compressed distributions. These differences underscore that senators face markedly different financial environments depending on the sectors from which they receive support.

This heterogeneity motivates the empirical strategy adopted in the paper. Rather than imposing a uniform effect of campaign finance across policy domains, the analysis allows the association between donations and speech to vary by sector. In the empirical models, sectoral donations therefore enter either individually, when examining topic-specific responsiveness, or jointly, in the form of donation shares, when assessing whether the effects of campaign finance are targeted or diffuse across policy domains. This dual specification makes it possible to capture both direct sector–topic relationships and broader patterns in how financial resources are associated with the allocation of legislative attention across multiple issues.

Control Variables

All empirical models include a set of control variables intended to capture alternative determinants of senators' speech content. Ideological position is measured using DW-NOMINATE scores, which capture persistent differences in legislators' policy preferences that may independently structure agenda priorities across policy domains.

The models also control for party affiliation, seniority in the Senate, and whether a senator is running for re-election in the subsequent electoral cycle, accounting for partisan differences, institutional experience, and electoral incentives that may shape communication strategies independently of campaign finance. In addition, indicators for partisan control of the Senate and for the party of the sitting President are included to capture broader political and institutional environments that structure legislative agendas. Finally, state fixed effects account for time-invariant geographic heterogeneity, such as differences in state-level economic structures or constituency interests.

Together, these controls allow the empirical strategy to isolate the association between campaign contributions and the thematic composition of senators' speeches while accounting for ideological, partisan, institutional, and geographic factors.

Dictionary

Rather than relying on unsupervised topic models, I adopt a dictionary-based approach to measure topic salience. Unsupervised methods such as Latent Dirichlet Allocation are well suited to discovering latent themes in large text corpora and have been widely used to explore ideological structure and rhetorical change. However, these approaches do not guarantee that inferred topics correspond to clearly defined and substantively meaningful policy domains. In contrast, a dictionary-based approach anchors textual measurement to ex ante categories that are directly comparable to the sectoral classification of campaign finance data, prioritizing interpretability and conceptual alignment over exploratory flexibility (Grimmer and Stewart 2013; Quinn et al. 2010). This feature is particularly important in analyses that seek to relate patterns of speech to external, non-textual data sources.

This choice is particularly appropriate given the focus of the analysis on sector–topic congruence. The goal is not to discover new or unexpected themes in Senate speeches, but to assess whether financial support from specific economic sectors is associated with increased attention to the corresponding policy domains. In this sense, the dictionary-based strategy is closely aligned with agenda-setting approaches that conceptualize political attention as a scarce resource strategically allocated across issues (Jones and Baumgartner 2005; Grimmer 2013). By fixing the set of topics ex ante, the approach allows for a direct test of whether variation in attention is systematically related to sector-specific contributions, rather than being driven by latent thematic structure.

The dictionary is constructed to mirror the classification used by OpenSecrets. For each of the thirteen sectors, I compiled a set of keywords and phrases designed to capture substantive references to sector-specific policy issues in Senate speeches. The construction process combines sectoral policy classifications, domain-specific terminology commonly used in legislative debates, and iterative inspection of the speech corpus to ensure that selected terms reflect actual usage rather than abstract policy labels.

In operational terms, the dictionary-based strategy also facilitates comparability across units and over time. Because the measurement relies on fixed keyword sets, changes in estimated topic salience can be interpreted as changes in the relative frequency with which senators invoke sector-relevant language, rather than changes in the composition of latent topics that may vary with corpus size, preprocessing choices, or model tuning. This stability is particularly valuable in settings where the substantive categories of interest are defined outside the text, here, the OpenSecrets sector classification, and where the empirical objective is to evaluate alignment between financial inputs and rhetorical outputs.

Moreover, constructing the dictionary at the sector level allows the analysis to distinguish between broad political language and policy-specific references. Many terms that might appear relevant at first glance (e.g., "jobs," "growth," or "industry") are too generic to provide meaningful evidence of sectoral responsiveness and would inflate salience measures across multiple domains. The conservative inclusion criteria therefore prioritize specificity, even when this reduces coverage, because the main inferential risk is false positive matching rather than omission. In this sense, the dictionary is designed to capture issue attention in a way that is directly interpretable in the regression models that follow, linking variation in measured salience to lagged sectoral contributions and ideological positioning.

An initial list of candidate terms was defined based on policy taxonomies and descriptive materials associated with the OpenSecrets classification and subsequently refined through corpus inspection. Particular care was taken to exclude overly generic terms and to retain keywords only when their dominant usage clearly corresponded to a specific policy domain. When ambiguity arose, more specific terms were favored, even at the cost of reducing coverage. This conservative strategy minimizes semantic overlap across sectors and reduces the risk that measured topic salience reflects generic political rhetoric rather than substantive engagement with sector-relevant issues (Grimmer and Stewart 2013).

The final dictionary consists of several dozen keywords per sector and is designed to balance coverage and precision. To illustrate the nature of the dictionary, representative examples are drawn from three policy domains with clearly defined and policy-oriented vocabularies. For the energy sector, the dictionary includes terms such as oil, natural gas, pipeline, renewable energy, and emissions. The health sector is represented by keywords such as Medicare, Medicaid, health insurance, pharmaceutical, drug, and hospital. The defense sector includes terms such as military, defense, weapon, national security, and Pentagon. These examples are intended for illustrative purposes only; the full dictionary is reported in Appendix A.

The dictionary is applied to the speech corpus to generate measures of topic salience. For each senator and electoral cycle, all speeches are concatenated into a single document, and topic salience is computed as the share of tokens matching the keywords associated with a given sector. Normalizing by the total number of tokens ensures comparability across senators with different levels of speech activity and across electoral cycles. The resulting measure captures agenda emphasis rather than evaluative tone or sentiment, consistent with agenda-setting frameworks that conceptualize speech as a form of legislative effort (Jones and Baumgartner 2005; Grimmer 2013).

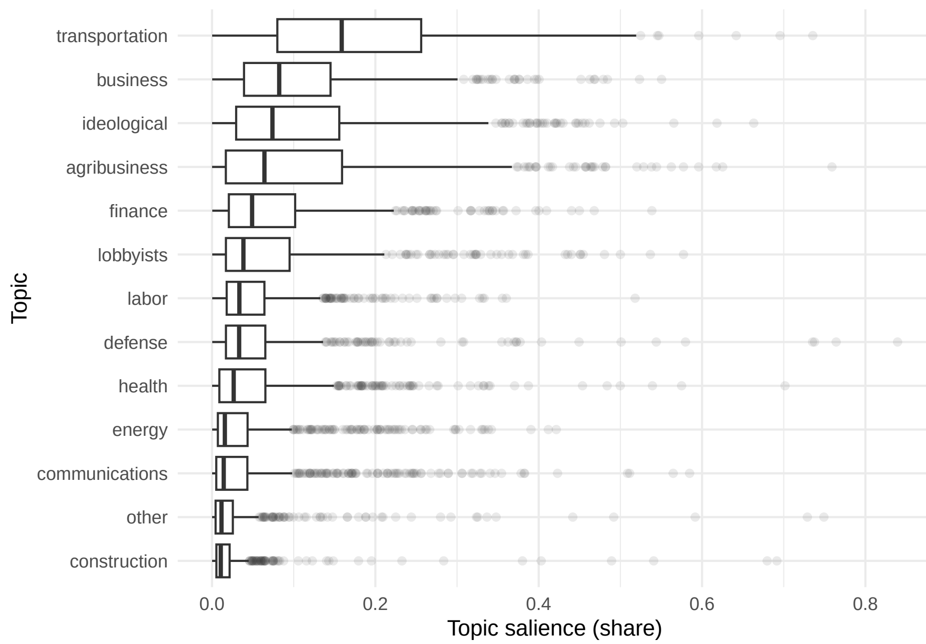

Figure 2: Topic salience distribution

Figure 2 reports the distribution of topic salience across policy domains. The figure highlights substantial heterogeneity in baseline attention across topics, with some domains exhibiting higher median salience and greater dispersion than others. This variation underscores the importance of accounting for topic-specific baselines in agenda-setting analyses and motivates the use of regression models that allow both ideological positioning and campaign contributions to explain deviations from these baseline levels of attention (Quinn et al. 2010; Jones and Baumgartner 2005).

The dictionary-based approach offers several advantages. It ensures a direct and transparent correspondence between sector-specific contributions and measured speech content, facilitating interpretation of the empirical results, and allows for straightforward replication and extension. At the same time, dictionary-based measures depend on the quality of the underlying keyword lists and do not capture framing differences within topics or contextual references. These limitations are mitigated by aggregating speeches at the senator–cycle level and by adopting conservative inclusion criteria during dictionary construction. While the approach does not capture all dimensions of political communication, it provides a robust and interpretable measure of thematic emphasis that is well suited to the research question of this paper.

Results

This section presents the empirical results of the analysis linking sector-specific campaign contributions to the thematic composition of U.S. Senate floor speeches. I proceed in three steps, corresponding to the hypotheses outlined in the previous section. I first examine whether campaign contributions from a given economic sector are associated with increased attention to the corresponding policy topic in senators' speeches (H1). I then assess the role of ideology in structuring agenda emphasis (H2), and finally evaluate whether the relationship between donations and speech is topic-specific rather than diffuse across policy domains (H3). Throughout the analysis, I report estimates from linear regressions at the senator–cycle level and include a rich set of controls capturing ideological, electoral, and institutional factors.

Hypothesis 1 posits that senators who receive higher campaign contributions from a given economic sector devote greater attention to the corresponding policy topic in their legislative speech. To test this hypothesis, I estimate a series of regressions in which the dependent variable is the relative salience of a given topic in a senator's floor speeches during an electoral cycle, and the main explanatory variable is the logarithm of lagged campaign contributions from the corresponding sector. All specifications include controls for party affiliation, ideological positioning (DW-NOMINATE), seniority, reelection incentives, party control of the Senate, party of the president, and state fixed effects.

Figure 3 summarizes the estimated coefficients and 95 percent confidence intervals. Overall, the results provide partial but meaningful support for H1. For several policy domains, sector-specific contributions are positively and statistically significantly associated with increased topic salience in legislative speech.

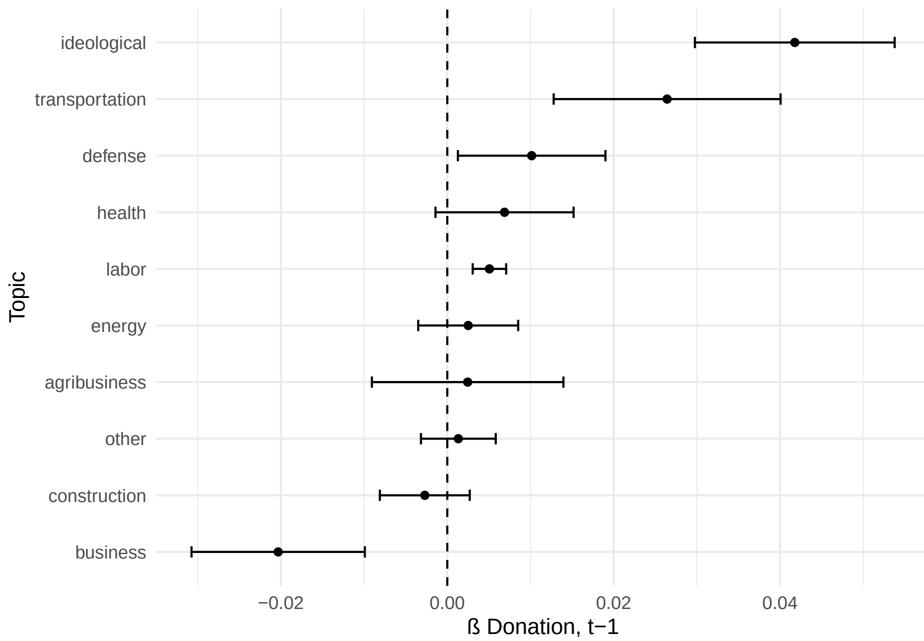

Figure 3: Estimated Effects of Sectoral Donations on Topic Shares

The strongest and most precisely estimated effect emerges for ideological organizations. Higher lagged contributions from ideological donors are associated with a substantial increase in the salience of ideological rhetoric, with the confidence interval clearly excluding zero. This effect is notably larger than those observed for other sectors, suggesting that ideologically motivated donors are particularly closely linked to senators' rhetorical emphasis on ideological themes.

Positive and statistically significant associations are also found for transportation, defense, and labor. Transportation-related contributions are associated with a sizable increase in topic salience, while defense and labor contributions display smaller but precisely estimated positive effects. These patterns are consistent with the expectation that organized interests seek to increase attention to policy domains of direct relevance to their sector.

For other sectors, including health, energy, agribusiness, and the residual "other" category, the estimated coefficients are positive but not statistically distinguishable from zero. While the direction of the association is broadly consistent with H1, the strength of the relationship varies considerably across sectors. In particular, sectors characterized by broader or more diffuse policy interests may exert weaker or less targeted influence on floor speech.

Finally, business contributions are negatively and significantly associated with business-related speech. This result suggests that higher business donations correspond to lower public emphasis on business topics, consistent with the interpretation that business interests may rely more heavily on backstage lobbying and committee activity rather than on public floor debate.

Taken together, these findings indicate that sector–topic congruence in legislative speech is present but uneven. While some sectors exhibit clear and economically meaningful associations between campaign contributions and topic salience, others show weaker or inconsistent patterns, consistent with an agenda-setting interpretation of campaign finance.

Hypothesis 2 examines whether legislators' ideological positions systematically structure baseline agenda emphasis in Senate floor speeches, independently of campaign contributions and other controls. Figure 4 reports the estimated relationship between DW-NOMINATE scores and topic salience across policy domains.

The results provide partial support for H2. Ideological positioning seems to be strongly associated with agenda emphasis in a subset of policy areas, while playing a limited role in others. In particular, more liberal senators devote greater attention to health and labor topics, consistent with long-standing ideological divisions over social policy and redistribution. These effects are among the most precisely estimated relationships in Figure 4.

By contrast, for several domains, including defense, energy, business, and agribusiness, the estimated associations between ideology and topic salience are small and statistically indistinguishable from zero. For these issues, baseline agenda emphasis does not appear to be systematically structured by ideological positioning once institutional and electoral factors are taken into account.

At the same time, Figure 4 shows that more conservative legislators place relatively greater emphasis on finance, transportation, ideological, and lobbying-related language. This pattern suggests that ideological differences shape not only positions on social policy, but also broader orientations toward economic organization, infrastructure, and

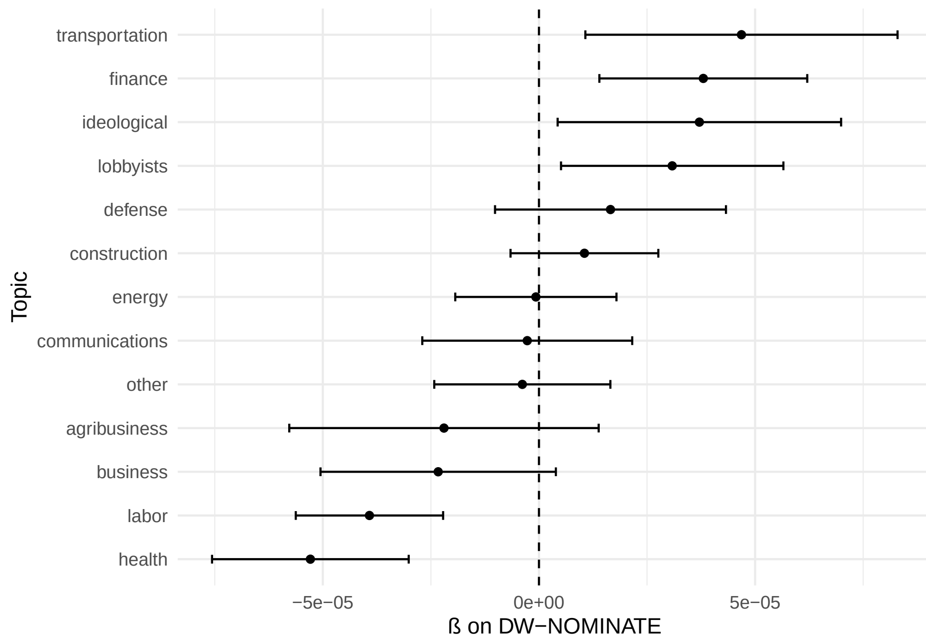

Figure 4: Ideology and baseline topic salience

political discourse.

Overall, the evidence indicates that ideology structures agenda priorities in a topic-specific rather than uniform manner. While ideological positioning predicts attention to some policy domains, it does not provide a general explanation for variation in speech across all topics, reinforcing the importance of accounting for both ideology and sector-specific influences in the analysis.

Hypothesis 3 evaluates whether the relationship between campaign contributions and legislative speech is topic-specific rather than diffuse across policy domains. To assess this hypothesis, I estimate regressions in which topic salience is regressed on the full vector of lagged sectoral donation shares. Figure 5 presents the resulting coefficient heatmap.

The heatmap reveals a pattern of topic-specific effects. For many policy domains, the largest and most statistically significant coefficients are concentrated along the diagonal of the matrix, indicating that increases in a sector's donation share are primarily associated with greater attention to the corresponding topic. Strong diagonal relationships emerge for labor, ideological, transportation, health, and agribusiness.

At the same time, relatively few large or significant off-diagonal coefficients are observed.

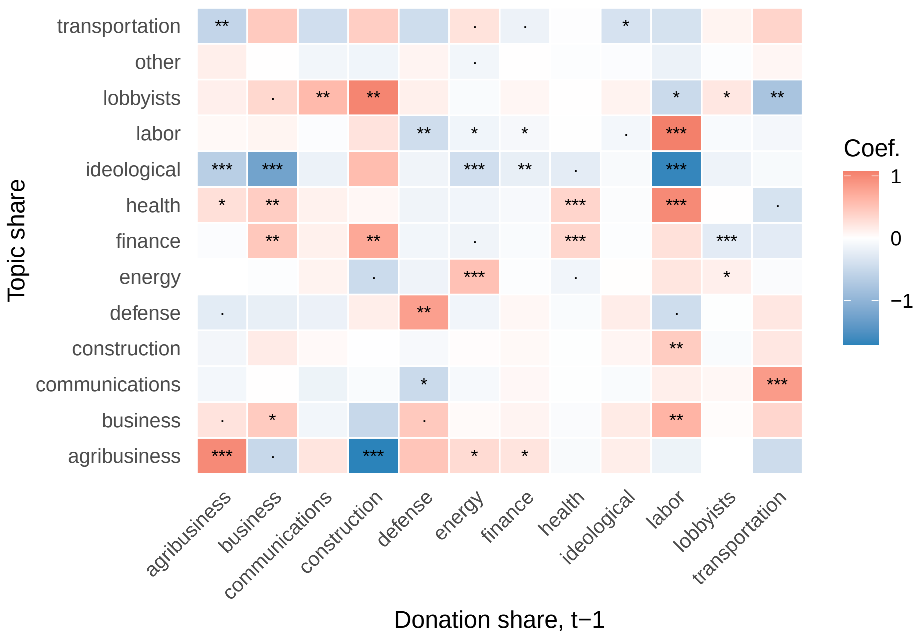

Figure 5: Sector–Topic Coefficient Matrix from Multivariate OLS Models

This suggests that, except for a few cases, sectoral donations do not generally translate into diffuse increases in attention across multiple policy areas, but instead exert targeted effects consistent with an agenda-setting mechanism.

Some departures from strict diagonal dominance are evident. In particular, lobbyist-related donations display positive associations with multiple topics, reflecting the cross-cutting nature of lobbying activity. Business-related donations also show weaker and more heterogeneous effects, reinforcing the interpretation suggested by the H1 results.

Importantly, topic specificity does not imply uniform effectiveness across all sectors. For some domains, diagonal effects are weak or insignificant, underscoring that the influence of campaign contributions on speech is heterogeneous across policy domains.

Overall, the multivariate analysis provides quite strong support for H3. Campaign contributions appear to shape legislative speech primarily through targeted, topic-specific agenda-setting dynamics rather than through broad rhetorical shifts.

Conclusion

This paper examines the relationship between campaign contributions and legislative speech in the U.S. Senate, asking whether sector-specific donations shape the thematic composition of floor speeches. Building on a growing literature that treats speech as a central dimension of legislative behavior, the analysis combines dictionary-based topic classification with senator–cycle–level regression models to study how money, ideology, and institutional context jointly structure agenda-setting in Congress. By focusing on speech rather than roll-call voting, the paper conceptualizes public communication as a measurable form of legislative effort and a potential channel through which organized interests shape political attention.

The results can be summarized briefly. First, campaign contributions are associated with increased attention to corresponding policy topics in legislative speech, providing partial support for a sector–topic congruence mechanism. While this relationship is not uniform across sectors, several domains exhibit positive and statistically significant associations between lagged contributions and topic salience, consistent with theories of lobbying that emphasize agenda-setting rather than direct vote-buying (Hall and Deardorff 2006; Witko et al. 2021). Second, legislators' ideological positioning is systematically related to baseline agenda emphasis for a subset of policy domains, underscoring the importance of accounting for ideology when studying the relationship between money and speech. Third, multivariate analyses indicate that the relationship between campaign contributions and legislative speech is predominantly topic-specific rather than diffuse across policy domains, with donations from a given sector primarily affecting attention to the corresponding policy domain. Taken together, these patterns suggest that donor influence operates through targeted shifts in attention layered on top of stable ideological differences in rhetorical priorities.

Overall, the analysis is subject to several limitations that warrant careful consideration. First, the results are observational and do not establish causal effects. Although the use of lagged campaign contributions and a rich set of controls mitigates concerns about simultaneity and reverse causality, unobserved factors, such as legislators' policy expertise, informal ties to interest groups, committee assignments, or strategic decisions about when to speak, may jointly influence both donations and speech.

Second, the dictionary-based measurement strategy necessarily abstracts from important dimensions of political communication. While this approach offers transparency and conceptual alignment between sectors and topics, it does not capture framing, tone, or within-topic rhetorical strategies, nor does it reliably identify implicit or highly contextual references to specific issues. Future research could complement the current strategy with supervised classification or embedding-based approaches and validate topic assignments through systematic human coding on a sample of speeches.

Third, the analysis focuses only on floor speeches, which represent only one arena of legislative communication. Many forms of influence may operate through committee hearings, bill drafting, amendments, or informal negotiations that are not captured in the dataset. If some sectors systematically prioritize backstage venues, estimates based on floor speech may understate their broader influence or mischaracterize how responsiveness is expressed across institutional arenas.

Finally, the aggregation of speeches at the senator–cycle level, while useful for capturing agenda priorities, necessarily smooths over short-term dynamics and strategic responses to specific events. Future work could exploit finer temporal variation, monthly or quarterly measures of speech attention, to examine whether contributions are followed by immediate rhetorical adjustments, whether effects concentrate around elections or salient policy debates, and whether agenda-setting dynamics differ in periods of heightened media attention.

Despite these limitations, the proposed analysis shows that campaign contributions, ideology, and institutional context are systematically related to legislative speech. By using a large dataset of political speeches, this paper highlights agenda-setting and rhetorical emphasis as important channels through which interest group influence may operate within U.S. institutions. More broadly, the findings underscore the value of studying legislative speech as a core component of democratic representation and provide a foundation for future research on feedback loops between money, communication, and political accountability.

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Appendix A: Dictionary Construction

This appendix reports the full list of keywords used to construct the sector-specific dictionary applied in the main analysis. The dictionary is organized by economic sector and mirrors the OpenSecrets classification. Keywords were selected using a conservative inclusion strategy to minimize semantic overlap across policy domains and ensure interpretability.

Sector	**Keywords**
Agribusiness	agricultural, agriculture, commodities, commodity, crop, farming, farm, farmer, food production, food supply, grain, livestock, meat, rural, soy, wheat
Business	business, chain, commerce, commercial, corporation, enterprise, entrepreneurship, firm, industry, market, private sector, retail
Communications	broadband, cyber, cybersecurity, data, digital, internet, media, networks, online, platform, social media, telecom, telecommunication
Construction	bridge, building, city, construction, contractor, housing, infrastructure, public work, real estate, road, urban development
Defense	air, armed, arm, army, combat, defense, drone, military, missile, national security, Pentagon, weapon, warfare, navy
Energy	climate, coal, drilling, electricity, emission, energy, fossil fuel, gas, green energy, natural gas, oil, pipeline, power grid, renewable energy
Finance	assets, bank, banking, capital, credit, debt, financial, finance, hedge funds, insurance, investment, loan, securities, stock, wall street
Health	biotech, care, disease, doctor, drug, healthcare, health insurance, hospitals, medical, medicine, medicaid, medicare, patient, pharmaceutical, Obamacare

Continued on next page

Sector	Keywords
Ideological	abortion, activism, beliefs, choice, constitution, constitutional, conservative, democracy, freedom, ideology, liberal, rights, values
Labor	bargaining, benefits, collective, employed, employment, job, labor, minimum wage, overtime, payroll, pension, union, wage, worker
Lobbyists	advocacy, campaign, compliance, courts, donations, ethics, influence, interest groups, lobbying, lobbyists, political action committees, gun
Other	art, charity, civic, culture, education, nonprofit, public service, religion, science, social services
Transportation	airlines, airports, aviation, cars, flight, freight, highways, logistics, ports, rail, shipping, transportation, travel, vehicles